

DECODELABS

Industrial Training Kit — Batch 2026

Project 3— queries Screenshot

Author	Thilani Senarath
Role	Data Analytics Intern
Organization	DecodeLabs
Dataset	E-commerce Orders — 1,200 rows
Tools	Python , Pandas ,Matplotlib , Seaborn
Date	May 27, 2026

1. EXECUTIVE SUMMARY

This report presents the findings of an Exploratory Data Analysis (EDA) conducted on the DecodeLabs e-commerce orders dataset, containing 1,200 orders across a 2.5-year period (January 2023 – June 2025). The objective was to uncover hidden patterns, distributions, trends, and outliers to generate actionable business intelligence.

Problem Statement

which products drive the most revenue, what the customer's preferred payment method is, why only a fraction of orders are successfully delivered, which marketing channels attract the most customers, and when peak sales periods occur.

Methodology

- Data loaded and explored using Pandas (`df.head()`, `df.info()`, `df.describe()`)
- Missing values identified and handled — 309 nulls in `CouponCode` filled with 'No Coupon'
- Grouped aggregations performed using `groupby()` for product and time-based trends
- Visualizations produced using Matplotlib (bar, line, pie) and Seaborn (countplot, boxplot)
- Outlier detection using the IQR (Interquartile Range) method
- Correlation matrix computed to identify variable relationships

2. DATASET OVERVIEW

The dataset contains 1,200 rows and 14 columns covering e-commerce order transactions.

Column	Data Type	Description
OrderID	Text	Unique order identifier
Date	Date	Order date 2023–2025
CustomerID	Text	Unique customer
Product	Category	7 product types
Quantity	Integer	1–5 units per order
UnitPrice	Decimal	\$11.39–\$699.93
ShippingAddress	Text	Delivery address
PaymentMethod	Category	5 payment types
OrderStatus	Category	5 status types
TrackingNumber	Text	Unique shipment code
ItemsInCart	Integer	1–10 cart items
CouponCode	Category	3 codes + 309 nulls
ReferralSource	Category	5 marketing channels
TotalPrice	Decimal	Final order value

Screenshots

File Edit View Query Database Server Tools Scripting Help

Navigator: project_3_mysql x

SCHEMAS

Filter objects

- PaymentMethod
- OrderStatus
- TrackingNumber
- ItemsInCart
- CouponCode
- ReferralSource
- TotalPrice
- Indexes
- Foreign Keys
- Triggers
- Views
- Stored Procedures
- Functions
- ecommerce_analysis
- sys

Administration Schemas

Information

Table: orders

Columns:

- OrderID varchar(10)
- Date date
- CustomerID varchar(10)
- Product varchar(50)
- Quantity int
- UnitPrice decimal(10,2)
- ShippingAddress varchar(100)
- PaymentMethod varchar(50)
- OrderStatus varchar(20)
- TrackingNumber varchar(50)

Result Grid

OrderID	Date	CustomerID	Product	Quantity	TotalPrice	OrderStatus
ORD200000	2023-01-04	C72649	Monitor	5	2853.10	Shipped
ORD200001	2024-08-23	C75739	Phone	2	302.70	Shipped
ORD200002	2024-02-27	C81728	Tablet	5	2753.40	Cancelled
ORD200003	2023-10-15	C33940	Chair	1	273.19	Returned
ORD200004	2025-05-08	C81840	Printer	4	2504.04	Delivered
ORD200005	2023-10-23	C37249	Phone	2	491.72	Shipped
ORD200006	2025-06-17	C83492	Laptop	1	664.42	Returned
ORD200007	2023-05-12	C41460	Monitor	5	747.75	Shipped
ORD200008	2025-04-02	C26817	Phone	2	268.56	Cancelled
ORD200009	2023-11-21	C31946	Desk	4	2037.52	Shipped
ORD200010	2023-12-29	C43443	Tablet	5	3129.85	Returned
ORD200011	2024-02-17	C93861	Monitor	3	147.42	Returned
ORD200012	2024-10-15	C38785	Monitor	2	361.00	Pending
ORD200013	2023-08-30	C88348	Laptop	3	604.47	Returned
ORD200014	2023-03-27	C98474	Tablet	2	786.66	Pending
ORD200015	2023-07-17	C39416	Printer	1	473.96	Delivered
ORD200016	2023-01-24	C28976	Printer	2	1067.62	Pending
ORD200017	2024-03-02	C60672	Tablet	1	423.40	Shipped
ORD200018	2025-03-26	C84329	Desk	1	431.44	Pending
ORD200019	2023-01-23	C14552	Monitor	5	1120.00	Returned

orders 1 x

Read Only

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Navigator: project_3_mysql x

SCHEMAS

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Result Grid

```
73
74 # Orders placed in the year 2024
75 SELECT OrderID, Date, Product, TotalPrice
76 FROM orders
77 WHERE Date BETWEEN '2024-01-01' AND '2024-12-31'
```

OrderID	Date	Product	TotalPrice
ORD200804	2024-01-01	Chair	709.50
ORD200931	2024-01-01	Printer	80.46
ORD200737	2024-01-03	Phone	2131.48
ORD200082	2024-01-05	Tablet	2260.80
ORD200548	2024-01-05	Printer	1278.36
ORD200399	2024-01-07	Tablet	2119.65
ORD201105	2024-01-08	Tablet	1427.07
ORD201164	2024-01-08	Tablet	1577.85
ORD200674	2024-01-09	Laptop	1435.00
ORD200712	2024-01-09	Tablet	136.80
ORD200781	2024-01-10	Phone	2621.30
ORD200936	2024-01-12	Chair	226.64
ORD200791	2024-01-13	Chair	1637.12
ORD200840	2024-01-14	Chair	1235.85
ORD200450	2024-01-15	Monitor	3075.50
ORD200476	2024-01-15	Phone	1027.18
ORD200624	2024-01-16	Tablet	1709.80
ORD200937	2024-01-16	Chair	597.06
ORD200113	2024-01-17	Laptop	950.40
ORD200410	2024-01-18	Tablet	1036.46

orders 2 x

Read Only

File Edit View Query Database Server Tools Scripting Help

Navigator: project_3_mysql x

SCHEMAS

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Administration Schemas

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Result Grid

```
120
121
122 # Top 10 highest value orders
123 SELECT OrderID, Product, TotalPrice
124 FROM orders
125 ORDER BY TotalPrice DESC
```

OrderID	Product	TotalPrice
ORD200789	Tablet	3456.40
ORD201122	Monitor	3390.95
ORD200632	Laptop	3390.80
ORD200469	Chair	3384.90
ORD200328	Tablet	3370.20
ORD200107	Printer	3353.75
ORD200326	Laptop	3352.40
ORD201065	Printer	3334.00
ORD201031	Phone	3322.55
ORD200463	Laptop	3313.90

orders 4 x

Output

Action Output

#	Time	Action	Message
2	10:47:54	SELECT OrderID, Date, Product, TotalPrice FROM orders WHERE Date BETWEEN 202...	459 row(s) returned
3	10:51:11	SELECT OrderID, Product, ReferralSource, TotalPrice FROM orders WHERE ReferralSour...	487 row(s) returned
4	10:51:26	SELECT OrderID, Product, TotalPrice FROM orders ORDER BY TotalPrice DESC LIMIT 10	10 row(s) returned

Object Info Session

File Edit View Query Database Server Tools Scripting Help

Navigator: project_3_mysql x

Limit to 1000 rows

140 # Sort by Product name, then TotalPrice descending

```

141 SELECT Product, OrderID, TotalPrice
142 FROM orders
143 ORDER BY Product ASC, TotalPrice DESC;
144

```

Result Grid

Product	OrderID	TotalPrice
Chair	ORD200469	3384.90
Chair	ORD200837	3277.75
Chair	ORD200527	3267.35
Chair	ORD200802	3223.20
Chair	ORD200241	3078.35
Chair	ORD200876	3044.55
Chair	ORD200222	2773.44
Chair	ORD200963	2651.70
Chair	ORD200287	2620.72
Chair	ORD200707	2600.16
Chair	ORD200977	2597.88
Chair	ORD200872	2517.36
Chair	ORD200567	2464.70
Chair	ORD200832	2408.55

Table: orders

Columns:

- OrderID varchar(20)
- Date date
- CustomerID varchar(20)
- Product varchar(20)
- Quantity int
- UnitPrice decimal(10,2)
- ShippingAddress varchar(100)
- PaymentMethod varchar(20)
- OrderStatus varchar(20)
- TrackingNumber varchar(20)

Object Info Session

Output

Action Output

#	Time	Action	Message
3	10:51:11	SELECT OrderID, Product, ReferralSource, TotalPrice FROM orders WHERE ReferralSource = 'ReferralSource'	487 row(s) returned
4	10:51:26	SELECT OrderID, Product, TotalPrice FROM orders ORDER BY TotalPrice DESC LIMIT 10	10 row(s) returned
5	11:07:27	SELECT Product, OrderID, TotalPrice FROM orders ORDER BY Product ASC, TotalPrice DESC	1000 row(s) returned

File Edit View Query Database Server Tools Scripting Help

Navigator: project_3_mysql x

Limit to 1000 rows

313

314 # Top 5 customers by total spending

```

315 SELECT CustomerID,
316        COUNT(*) AS TotalOrders,
317        ROUND(SUM(TotalPrice), 2) AS TotalSpent
318 FROM orders
319 GROUP BY CustomerID
320 ORDER BY TotalSpent DESC
321 LIMIT 5;

```

Result Grid

CustomerID	TotalOrders	TotalSpent
C38840	2	5723.23
C57276	1	3456.40
C67260	1	3390.80
C13877	1	3384.90
C18404	1	3370.20

Table: orders

Columns:

- OrderID varchar(20)
- Date date
- CustomerID varchar(20)
- Product varchar(20)
- Quantity int
- UnitPrice decimal(10,2)
- ShippingAddress varchar(100)
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- OrderStatus varchar(20)
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Object Info Session

Output

Action Output

#	Time	Action	Message
4	10:51:26	SELECT OrderID, Product, TotalPrice FROM orders ORDER BY TotalPrice DESC LIMIT 10	10 row(s) returned
5	11:07:27	SELECT Product, OrderID, TotalPrice FROM orders ORDER BY Product ASC, TotalPrice DESC	1000 row(s) returned
6	11:10:21	SELECT CustomerID, COUNT(*) AS TotalOrders, ROUND(SUM(TotalPrice), 2) AS TotalSpent FROM orders GROUP BY CustomerID ORDER BY TotalSpent DESC LIMIT 5	5 row(s) returned

File Edit View Query Database Server Tools Scripting Help

Navigator: project_3_mysql x

Limit to 1000 rows

303 # Revenue from coupon users vs non-coupon users

```

304 SELECT CASE
305        WHEN CouponCode IS NOT NULL AND CouponCode != '' THEN 'Used Coupon'
306        ELSE 'No Coupon'
307 END AS CouponUsage,
308 COUNT(*) AS TotalOrders,
309 ROUND(SUM(TotalPrice), 2) AS TotalRevenue,
310 ROUND(AVG(TotalPrice), 2) AS AvgOrderValue
311 FROM orders
312 GROUP BY CouponUsage;
313
314 # Top 5 customers by total spending

```

Result Grid

CouponUsage	TotalOrders	TotalRevenue	AvgOrderValue
Used Coupon	891	942360.55	1057.64
No Coupon	309	322401.41	1043.37

Table: orders

Columns:

- OrderID varchar(20)
- Date date
- CustomerID varchar(20)
- Product varchar(20)
- Quantity int
- UnitPrice decimal(10,2)
- ShippingAddress varchar(100)
- PaymentMethod varchar(20)
- OrderStatus varchar(20)
- TrackingNumber varchar(20)

Object Info Session

Output

Action Output

#	Time	Action	Message
5	11:07:27	SELECT Product, OrderID, TotalPrice FROM orders ORDER BY Product ASC, TotalPrice DESC	1000 row(s) returned
6	11:10:21	SELECT CustomerID, COUNT(*) AS TotalOrders, ROUND(SUM(TotalPrice), 2) AS TotalSpent FROM orders GROUP BY CustomerID ORDER BY TotalSpent DESC LIMIT 5	5 row(s) returned
7	11:12:13	SELECT CASE WHEN CouponCode IS NOT NULL AND CouponCode != '' THEN 'Used Coupon' ELSE 'No Coupon' END AS CouponUsage, COUNT(*) AS TotalOrders, ROUND(SUM(TotalPrice), 2) AS TotalRevenue, ROUND(AVG(TotalPrice), 2) AS AvgOrderValue FROM orders GROUP BY CouponUsage	2 row(s) returned